CHAPTER 10

Energetics of Nuclear Reactions

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Reactions between an atomic nucleus and another particle are called nuclear reactions. In some such reactions, the nucleus is unchanged (*elastic scattering*), in others the original nucleus is excited to a higher energy state (*inelastic scattering*); in a third class new nuclei are formed (*nuclear transmutations*). Spontaneous nuclear transformations, which are involved in the radioactive decay of unstable nuclei, have been discussed in Chapter 5. In this chapter the emphasis is on the mass and energy relationships when a projectile interacts with a nucleus.

10.1. CONSERVATION LAWS IN NUCLEAR REACTIONS

We have previously discussed a number of laws governing nuclear processes. A summary of these laws provides us with a picture of the mechanics of nuclear interactions occurring below ~100 MeV/A. At higher energies the mass number may not be conserved.

Let us begin by considering the simple process shown in Figure 10.1, in which a *projectile*, indicated by subscript 1, collides with a *target* atom, subscript 2, forming an intermediate system, subscript i. The intermediate system splits into the products, which are designated by subscripts 3 and 4. We call this a *nuclear reaction*. The velocity of the projectile v_1 is greater than zero, but the velocity of the target atom v_2 is made to be zero by using the target nucleus as the origin of the system of reference.

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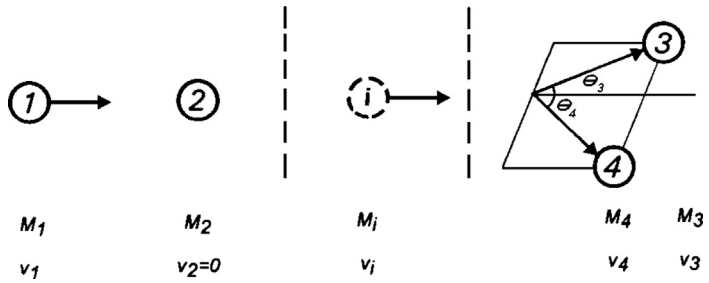


Figure 10.1 Schematic picture of a nuclear reaction.

All the *conservation laws* derived in §5.2 are applicable to nuclear reactions. These are:

(a) the conservation of total energy:	$\Delta E = 0$	(5.4)
(b) the conservation of linear momentum:	$\Delta p = 0$	(5.6)
(c) the conservation of total charge:	$\Delta Z = 0$	(5.8)
(d) the conservation of mass number:	$\Delta A = 0$	(5.9)
(e) the conservation of spin:	$\Delta I = 0$	(5.11)

"Conservation of total energy" means that the total energy of the products must equal the total energy of the reactants, i.e. $E_{\text{products}} - E_{\text{reactants}} = \Delta E = 0$. For (5.6) it should be remembered that linear momentum is a vector property; thus

$$p_1 + 0 = p_3 \cos \theta_3 + p_4 \cos \theta_4 \tag{10.1}$$

$$p_3 \sin \theta_3 = p_4 \sin \theta_4 \tag{10.2}$$

where θ_3 and θ_4 are the respective emission angles, see Fig. 10.1. As pointed out earlier, some of these conservation laws (e.g. (d) and (e)) are not always obeyed in high energy reactions in which new elementary particles may be formed. The kinetic energy equation $E_{\text{kin}} = \frac{1}{2}mv^2$ (3.6) was deduced by Newton in 1687 assuming that the mass of the particle was independent of its velocity. We have seen that this is not true at high particle velocities (cf. Fig. 5.2). For example the relativistic mass increase of a bombarding proton is about 1% at a kinetic energy of 10 MeV. In all dynamic relations involving moving particles, the relativistic mass must be used. This is particularly important in Ch. 16, where we discuss the effect of the acceleration of charged particles to high energies in nuclear particle accelerators.

10.2. THE MASS ENERGY

As for radioactive decay, the energy of a nuclear reaction is given by its Q-value (4.10, 5.14):

$$Q\,(\text{MeV}) = -931.5\,\Delta\,M^0(\text{u}) \tag{10.3}$$